



**CALICUT UNIVERSITY – FOUR-YEAR
UNDERGRADUATE PROGRAMME (CU-FYUGP)**

BSc CHEMISTRY

Programme	B.Sc Chemistry				
Course Title	INORGANIC CHEMISTRY I				
Type of Course	MAJOR/MINOR				
Semester	I				
Academic Level	100-199				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	3	-	2	75
Pre-requisites	Scope of chemistry, Interdisciplinary areas involving chemistry. Fundamentals of periodic properties of elements, Atoms and molecules, Need for chemical bonding and its types, Awareness on nature of experiments and health risk, hazard associated with chemicals, Mole concept				
Course Summary	This course explores the importance of chemistry as a central discipline of science. It introduces the periodic properties of elements, concept of chemical bonding and explanation of inorganic molecular structure using hybridization and MO theory. A few basic topics of the emerging area of Nanochemistry are also introduced in this course. The basic laboratory safety, concepts in volumetric analysis and related practical experiments are also covered.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Understand the role of chemistry in science and scientific research with emphasis on analytical data evaluation	U	C	Instructor- created exams/ Quizzes/Assignments
CO2	Conceptualize and predict chemical bonding, molecular structures using	An	P	Instructor- created exams/ Quizzes/assignments

	dipole moment, hybridisation, and MO Theory			
CO3	Develop a basic understanding of the extraordinary properties of nanomaterials and its applications.	U	C	Instructor- created exams/ Quizzes/Assignments
CO4	Apply the concepts of lab safety measurements and volumetric analysis	Ap	M	Instructor- created exams/ Assignments/problem solving
CO5	Enable students to develop analytical skills in inorganic quantitative volumetric analysis.	Ap	P	Group work /Viva Voce// Observation of practical skill
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (45+30)	Marks
I	CHEMISTRY AS A SCIENCE DISCIPLINE & SCIENTIFIC ANALYSIS		8	17
	1	Science- Chemistry as a branch of science, History of chemistry, Involvement of chemistry in daily life (Mention only)	1	
	2	Introduction to analytical chemistry, Classification of analytical methods: Qualitative and Quantitative analysis (Mention with examples)	1	
	3	Treatment of analytical data - Significant figures – Accuracy – Precision – Methods of representing Accuracy, Absolute error, Relative error, Types of errors, Constant errors, Proportional errors, Correction of determinate errors	3	
	4	Methods of representing Precision –Mean, Average deviation, Standard deviation, Relative standard deviation, Coefficient of variation, Variance, Rejection of a result: Q test, Methods of least squares	3	
II	CHEMICAL BONDING AND MOLECULAR STRUCTURE		17	38
	5	Periodic Properties and their Periodic Trends: (a) Atomic and Ionic radius (include isoelectronic species in discussion) (b) Ionisation energy: (c) Electron	2	

		affinity (d) Electronegativity (Pauling, Mulliken Allred & Rochow scales).		
	6	Classification of bonds: Ionic bond - Definition, Factors affecting the formation of ionic bond. Characteristics of ionic compounds. Lattice energy	1	
	7	Born Haber cycle - Born Lande equation (derivation not needed) - Covalent –(Mention polar and non polar compounds) and Coordinate bond	2	
	8	Dipole moment and its applications: (Prediction of linearity and symmetry of polyatomic molecules, Prediction of position of substituents in aromatic compounds, Measurement of bond angle)	2	
	9	Covalent Bond, Lewis concept of covalent bond, Atomic orbital overlap, Concept of covalency, Variable covalency and Maximum covalency	2	
	10	Prediction of Covalent character in ionic bond using Fajans rule. Prediction of Ionic character in Covalent bond using Hannary Smidth equation.	1	
	11	Structure of molecules by the concept of Hybridisation: NO_3^- , CO_3^{2-} , SO_4^{2-} , IF_7 , XeO_3 , XeO_4 , XeF_2 , XeF_4 , XeF_6 , ClF_3 , BrF_5 , SF_4	3	
	12	Introductory MO Theory: Homoatomic molecules in N_2 and O_2 and their ions (comparison of bond order, bond length and stability), MO Theory: Heteroatomic molecules like NO, CO, HCl, HF, LiF.	4	
III	INTRODUCTION TO NANOMATERIALS		10	21
	13	Definition of Nanomaterials, Historical revolution of Nanochemistry, Nanochemistry and Nanotechnology, Classification of nanostructures based on electron confinement (0D, 1D and 2D)	2	
	14	Synthesis of Nanomaterials: Bottom Up and Top down approaches (Elementary idea with examples)	1	
	15	Metal nanoparticles (gold and silver nanoparticles), Semiconductor nanoparticles (CdS and CdSe nanoparticles), Metal oxide nanoparticles (zinc oxide, iron oxide, silica and titania nanoparticles), Nanocomposites, Nanoceramics (Definition with examples), Carbon Based Nanomaterials: Graphene, Carbon Nanotubes, Fullerenes, Carbon dots (elementary idea only)	2	
	16	Characteristics of Nanomaterials: Surface area to volume ratio and its significance, Novel properties of Nanomaterials, Size dependent optical (surface	3	

		plasmon resonance), Electronic, Mechanical, magnetic and catalytic properties (No deep discussion is needed)		
	17	Applications of nanomaterials: Electronics (Batteries, Solar cell), Biomedical (Drug Delivery) and Environmental based applications (Water Purification, Dye Removal) (General idea only)	2	
IV	FUNDAMENTALS OF ANALYTICAL CHEMISTRY		10	22
	18	Lab safety measurements: Awareness of material safety data sheet (MSDS), Safe storage and handling of hazardous chemicals, Simple first aids; Electric shocks, fire, Cut by glass and inhalation of poisonous gas.	2	
	19	Accidents due to acids and alkalis, Burns due to phenol and bromine, Disposal of waste chemicals, Disposal of sodium and broken mercury thermometer, -R and S phrases (elementary idea only), Personal protective Equipment (PPE)	1	
	20	Mole concept - Equivalent mass - Methods of expressing concentration: Weight percentage, molality, molarity, normality, mole fraction, ppm and millimoles - Numerical Problems related to basic concepts.	2	
	21	Volumetric Analysis: Introduction - Primary and secondary standards – Standard solutions – Theory of titrations involving acids and bases, Permanganometry, Dichrometry, Iodometry, Iodimetry Precipitation and Complexometric titrations.	3	
	22	Indicators: Theory of acid-base, redox, adsorption and complexometric indicators. Double burette method of titration: Principle and advantages.	2	
V	INORGANIC CHEMISTRY PRACTICAL I- VOLUMETRIC ANALYSIS		30	
	1	General Instructions: Use a safety coat, gloves, shoes and goggles in the laboratory. For weighing electronic balance must be used. Double burette titration method may be used for titrations. Standard solution must be prepared by the student. A minimum of 7 experiments must be done from Section B and C.		

		Section D is open-ended and the experiments can be selected by the teacher SECTION A Importance of lab safety – Burns, Eye accidents, Cuts, Gas poisoning, Electric shocks, Treatment of fires, Precautions and Preventive measures. Weighing using electronic balance, Preparation of standard solutions. SECTION B Neutralization Titrations 1. Acidimetry and Alkalimetry: Strong acid Vs Strong base 2. Acidimetry and Alkalimetry: Strong acid Vs Weak base SECTION C Redox Titrations 1. Permanganometry: Estimation of $\text{Fe}^{2+}/\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$/Mohr's salt 2. Permanganometry: Estimation of Oxalic acid 3. Permanganometry: Estimation of Calcium using std KMnO_4 4. Dichrometry: Estimation of $\text{Fe}^{2+}/\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$/Mohr's salt 5. Dichrometry: Estimation of Ferric iron 6. Iodometry and Iodimetry: Estimation of Copper 7. Iodometry and Iodimetry: Estimation of Iodine SECTION D Open Ended (Any two experiments are to be conducted. may be selected from the below list or the teacher can select related experiments) 1. Determination of acetic acid content in vinegar by titration with NaOH.		
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		2. Determination of alkali content in antacid tablets by titration with HCl.		
		3. Determination of available chlorine in bleaching Powder.		
		4. Estimation of Cu in Brass		

References

1. C. N. R. Rao, *Understanding Chemistry*, Universities Press India Ltd, Hyderabad, 1999
2. George Gamow, *One, Two, Three...Infinity: Facts and Speculations of Science*, Dover Publications, 1988.
3. *Resonance – Journal of Science Education*, Indian Academy of Sciences.
4. *Nature Chemistry*, Nature Publishing Group.
5. J. Mendham, R. C. Denney, J. D. Barnes, M. Thomas, *Vogel's Text Book of Quantitative Chemical Analysis*, 6th Edn., Pearson Education, Noida, 2013.
6. D. A. Skoog, D. M. West, F. J. Holler, S. R. Crouch, *Fundamentals of Analytical Chemistry*, 8th Edn., Brooks/Cole, Thomson Learning, Inc., USA, 2004.
7. B. R. Puri, L. R. Sharma, K. C. Kalia, *Principles of Inorganic Chemistry*, 31st Edn., Milestone Publishers and Distributors, New Delhi, 2013.
8. Satya Prakash, *Advanced Inorganic Chemistry*, Vol. 1, 5th Edn., S. Chand and Sons, New Delhi, 2012.
9. J. D. Lee, *Concise Inorganic Chemistry*, 5th Edn., Oxford University Press, New Delhi, 2008
10. W. U. Malik, G. D. Tuli, R. D. Madan, *Selected Topics in Inorganic Chemistry*, S. Chand and Co., New Delhi, 2010.
11. J. E. Huheey, E. A. Keitler, R. L. Keitler, *Inorganic Chemistry – Principles of Structure and Reactivity*, 4th Edn., Pearson Education, New Delhi, 2013.
12. C.N.R., Rao, A. Müller, and A.K. Cheetham, (Eds.), "Chemistry of Nanomaterials", Wiley – VCH. 2005
13. T., Pradeep, *A Textbook of Nanoscience and Nanotechnology*, McGrawhill, New Delhi, 2012.
14. M. A. Shah, Tokeer Ahmad, *Principles of Nanoscience and Nanotechnology*, Narosa Publishing House, New Delhi, 2010.
15. V. S. Muralidharan, A. Subramania, *Nano Science and Technology*, CRC Press, London.
16. R. H. Hill, D. Finster, *Laboratory Safety for Chemistry Students*, 1st Edn., Wiley, Hoboken, NJ, 2010.

Further Reading

1. H. Collins, T. Pinch, *The Golem: What Everyone Should Know about Science*, Cambridge University Press, Cambridge, 1993.
2. C.R. Kothari, *Research Methodology: Methods and Techniques*, 2nd Revised Edition,

New Age International Publishers, New Delhi, 2004.

3. <http://www.vlab.co.in>
4. <http://nptel.iitm.ac.in>
5. D. F. Shriver, P. W. Atkins, *Inorganic Chemistry*, 5rd Edn., Oxford University Press, New York, 2010.
6. M. C. Day, J. Selbin, *Theoretical Inorganic Chemistry*, East West Press, New Delhi, 2002.
7. G. L. Miessler, D A. Tarr, *Inorganic Chemistry*, Pearson, 2010
8. K.J. Klabunde (Ed.), "Nanoscale Materials in Chemistry", John Wiley & Sons Inc. 2001
9. G., Schmidt, *Nanoparticles: From theory to applications* –Wiley Weinheim 2004.

Mapping of COs with PSOs and POs:

	PSO 1	PSO 2	PSO 3	PSO 4	PSO 5	PSO 6	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7
C O 1	3			2	1	1	3				3	1	1
C O 2	2	2					2				2		1
C O 3	2		1	2	2	3	2			1	2	1	2
C O 4			3		2	2	2		1		1	1	1
C O 5			3		2	3	3		1		2	1	2

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Assignment/ Discussion / Seminar
- Midterm Exam
- Practical exam (20%)
- Final Exam (70%)

Mapping of COs to Assessment Rubrics:

	Internal Exam	Assignment/ Viva/ Seminar	Practical skill evaluation	End Semester Examinations
CO 1	✓	✓		✓
CO 2	✓	✓		✓
CO 3	✓	✓		✓
CO 4	✓	✓		✓
CO 5	✓	✓	✓	